

# EGE UMUT CAN

Department: Economics, Finance & Quantitative Data Science

Istanbul, Türkiye | FMV Isik High School, Ispartakule Campus (Class of 2027) | Portfolio: ege211.github.io/macroeconomic-data-lab | GitHub: github.com/ege211

## EDUCATION & QUALIFICATIONS

### FMV Ozel Ispartakule Isik High School (FMV Isik Schools), Istanbul, Türkiye

Sept 2023 – June 2027

- **Academic Standing:** Cumulative GPA: **90.0 / 100.0** (9th–11th Grade High Honour / Honour Roll Standing)
- **Advanced Placement (AP) Coursework:** AP Macroeconomics, AP Microeconomics, AP Calculus, AP Psychology
- **Academic Focus:** Monetary Policy Credibility, Inflation Persistence, Market Microstructure, Financial Econometrics

## SUMMER SCHOOLS, INTERNATIONAL CERTIFICATIONS & ONLINE COURSES

### Docklands Academy London — Intensive English & UK Summer School, London, UK

9th–10th Grade, 20 hrs/wk (June 2024 – 2025)

Student; Accredited by the British Council & British Accreditation Council (BAC), Member of English UK & Study UK

- **CEFR B2, B2+ & C1 Advanced Certifications:** Completed intensive academic English training in London (including the 10–21 June 2024 residential module), earning official progression certificates across **CEFR B2 (Upper-Intermediate)**, **B2+**, and **C1 (Advanced Proficient User)** levels.
- **Academic & Field Immersion:** Participated in analytical writing seminars and cross-cultural debates in London's Canary Wharf / Docklands district, while conducting observational field studies on UK hospitality service-profit chains.

### Bank of America — Global Markets Sales & Trading & Private Bank Simulations (Forage)

11th Grade, 6 hrs/wk (June – Sept 2026)

Analyst Participant; Verified Completion IDs: 8bd49n4kfwMYweRGZ (Global Markets) & 6ab97c46a90bc149453c6cc5 (Private Bank)

- **Global Markets & Private Bank:** Authored a *Market Insight Report*, a *Strategic Recommendation Proposal* (Summit Investments), and a *Process Improvement Proposal* (Apex Global Investments) on live market dynamics, workflow automation, and multi-generational Private Bank wealth allocation.

### Goldman Sachs — Risk Analysis Job Simulation (Forage)

11th Grade, 4 hrs/wk (June 2026)

Analyst Participant; Verified Completion ID: eDZtiYexR4uGgD9Br

- **Credit & Real Estate Risk Profiling:** Evaluated institutional client balance-sheet profiles and commercial real-estate investment exposures under downside macroeconomic stress scenarios.

## RESEARCH PROJECTS & ECONOMETRIC WORKING PAPERS

### Comparative Macroeconomic Data Lab (GEA) — Founder & Lead Student Researcher

10th–11th Grade, 8 hrs/wk, 36 wks/yr (2025–Present)

Founder & Developer; Live Web Lab: ege211.github.io/macroeconomic-data-lab | GitHub: github.com/ege211/macroeconomic-data-lab

- **Live API & Dual Simulators:** Engineered an interactive web laboratory retrieving live **World Bank API v2** feeds (GDP growth, CPI inflation) across 5 economies (2019–2023), featuring a *2024–2025 Cross-Country Scenario Simulator* and a *WP-01 AR(1) Inflation Persistence & Shock Half-Life Simulator* ( $\beta = 0.72$ , half-life  $t_{1/2} = 2.11$  years vs. 0.58 years in anchored regimes).
- **10-Paper Research Corpus:** Authored and published 10 full-text working papers, empirical essays, and methodological research notes across 4 thematic pillars, accompanied by a 6-work theoretical reading list (Acemoglu & Robinson, Gali & Gertler, Taylor, Piketty, SPK III-52.1, Porter).

### A Machine-Learning-Based Early Warning Framework for Systemic Banking Crises

11th Grade, 10 hrs/wk, 12 wks/yr (2026)

- **Investigator:** Evaluated macro-financial indicators across a balanced panel of **76 World Bank economies (1990–2025; 684 out-of-sample evaluations)** using penalized Logistic Regression and Random Forest within a leak-free expanding-window validation pipeline ( $T = 2000–2008$ ; 2.78% crisis base rate).
- **Key Finding:** Demonstrated that adding credit-to-GDP leverage and external deficits more than doubled pooled recall to **47.37%** (vs. 21.05% baseline), while diagnosing an out-of-sample ranking breakdown (**ROC-AUC = 0.2301**) outside the 2007 pre-GFC cluster.

### AI-Assisted Equity Valuation: A Walk-Forward Study on SEC Form 10-K Filings

11th Grade, 10 hrs/wk, 10 wks/yr (2026)

- **Investigator:** Extracted 12 linguistic dimensions across **30 S&P large-cap firms (2014–2024)** via a five-tier SEC EDGAR XBRL normalization cascade linked to a deterministic DCF/Reverse DCF valuation bridge (*valuation\_bridge\_v1.0*); proved out-of-sample that 10-K text signals yield **zero incremental predictive power** over trailing fundamentals ( $p = 0.2335$ ), with unconstrained text inducing **-13.14% degradation**.

### Strategic Capital Allocation in Investment Banking: G-SIBs vs. Advisory Boutiques

11th Grade, 8 hrs/wk, 10 wks/yr (2026)

- **Investigator:** Hand-collected **540 audited observations across 40 SEC Form 10-K filings (2020–2024)** disclosing **\$233.2B in shareholder distributions** (\$113.8B dividends, \$119.3B repurchases); identified near-zero ROE correlation in BHCs ( $r = -0.049$ ) vs. negative margin correlation in boutiques ( $r = -0.403$ ) driven by Basel III **CET1 buffers (13.1%–17.4%)** vs. talent-retention liquidity.

### Inflation Persistence (WP-01) & Arsenal FC Longitudinal Sports Finance Study

11th Grade, 6 hrs/wk, 16 wks/yr (2026)

- **Cross-Country Panel (WP-01):** Estimated dynamic panel regressions across **45 economies (2000–2023)**, identifying emerging-market inflation persistence of  **$\beta = 0.72$**  and a 2–4 quarter policy transmission lag ( $\gamma < 0$ ).
- **Arsenal FC Study (2015/16–2024/25):** Integrated 10 seasons of audited accounts, **128 transfers**, and *Understat xG/xPTS*; showed transfer spend correlates with tactical process ( **$xGD\ r = +0.5982$ ;  $xPTS\ r = +0.6189$** ) rather than noisy match points (PPG  $r = +0.3911$ ), while refuting single-club wage dominance ( $r = +0.0516$ ).

### Macroeconomic Policy, Market Microstructure & Strategy Essays (Published on Data Lab)

10th–11th Grade (2025–2026)

- **Borsa Istanbul — Fragility of Liquidity (2026):** Analyzed retail equity inflows (2022–2023), the May 2026 liquidity contraction, mutual fund redemption spirals under **SPK Communiqué III-52.1**, and corporate bond (*tahvil*) risk pricing.
- **Monetary & Political Economy Essays:** Authored *Exchange Rate Pass-Through in Emerging Markets* (ERPT 0.38 vs. 0.09), *The Credibility Gap*, *The Lira Doesn't Lie*, *The Paradox of Petrowealth* (Venezuela vs. Norway/GCC), *Contested Terrain* (Card & Krueger, Piketty  $r > g$ ), and the *Inditex (Zara) VRIO/Five Forces Case Study*.

WORK EXPERIENCE & CORPORATE INTERNSHIPS

- TEB Arval (BNP Paribas Group), Gayrettepe HQ, Istanbul — Fleet Sales, Operations & HR Intern** 11th Grade, 40 hrs/wk (19–26 Jan 2026)  
*Intern; Supervised by Human Resources & Fleet Sales Organization Department (Official Reference Letter Awarded)*
- Observed end-to-end B2B corporate fleet leasing workflows, organizational process architecture, and corporate client relationship management at TEB Arval (BNP Paribas joint venture), analyzing administrative capacity bottlenecks and cross-departmental KPI coordination.
- Haber Tekstil Dis Ticaret (Bohoo Textile Export Supplier), Istanbul — Export Department Intern** 11th Grade, 35 hrs/wk (Jan – Feb 2026)  
*Intern & Field Researcher; Foreign Trade / Export Department (Official Completion Certificate Awarded)*
- Analyzed B2B international apparel export workflows, customs documentation, and customer value propositions (CVP), documenting how relationship trust and 2–3 week lead-time guarantees insulate mid-market Turkish exporters from pure price competition.

COMMUNITY INVOLVEMENT & SOCIAL RESPONSIBILITY PROJECTS

- Sister School Annual Book Drive & Educational Equity Initiative (Kardes Okul Projesi), Istanbul** 9th–11th Grade, 4 hrs/wk, 8 wks/yr (2023–Present)  
*Organizer & Volunteer; Annual School-Wide Campaign at FMV Isik High School*
- Spearheaded recurring annual book collection campaigns across 9th, 10th, and 11th grades to expand library access and educational equity for students at under-resourced sister schools (*Kardes Okul*) across Türkiye.
  - Coordinated student outreach, categorized donated textbooks, reference works, and literature by grade level, and managed packaging and shipment logistics each academic year.

SPORTS, MUSIC & EXTRACURRICULAR ACHIEVEMENTS

- FMV Isik High School Varsity Football (Soccer) Team — 3rd Place Bronze Medalist** 9th–11th Grade, 4 hrs/wk, 32 wks/yr (2023–Present)
- **Inter-District Championship Medalist:** Competed as a core member of the FMV Isik High School football team, winning the **3rd Place Bronze Medal** in the Istanbul Inter-District High School Football Tournament (*Ilceler Arasi Futbol Turnuvasi*).
  - Cultivated tactical teamwork, physical endurance, and composure under competitive pressure while balancing an intensive AP and research workload.
- Classical & Contemporary Piano Performance** 9th–11th Grade, 2 hrs/wk, 40 wks/yr (Ongoing)
- Practiced solo piano performance, sight-reading, and harmonic interpretation, developing long-term artistic discipline, focus, and creative expression alongside quantitative studies.

SKILLS, LANGUAGES & PERSONAL DETAILS

- **Quantitative & Technical Tooling:** Python (*pandas, scikit-learn, numpy, statsmodels, pytest*), SQL / DuckDB, SEC EDGAR XBRL Parsing, Deterministic & Reverse DCF Modeling, World Bank API v2, Git / GitHub, HTML5 / CSS3 / JavaScript (*Chart.js*).
- **Languages:** Turkish (Native), English (**CEFR C1 Advanced** — Docklands Academy London B2, B2+ & C1 Progression Certificates).
- **Hobbies & Interests:** Competitive football, playing the piano, macroeconomic history, sports analytics, financial market microstructure.
- **Personal Details:** Date of Birth: January 17, 2008 | Place of Birth: Istanbul, Türkiye | Nationality: Turkish.